

Multi-Tranche Arbitrage Deposit Planning across Dual Banking Systems over a General Time Horizon

Profit Decomposition, Stochastic Modelling,
Linear-Programming Allocation, and
Machine-Learning Rate Forecasting

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Abstract

We generalise the two-deposit, unit-horizon arbitrage deposit plan to a setting with an arbitrary finite number n of deposits $D_1, \dots, D_n$ placed at scheduled times $0 = s_1 < s_2 < \dots < s_n < \mathcal{T}$ within a banking system operating over the general time horizon $[0, \mathcal{T}]$, where $\mathcal{T} > 0$. Bank A quotes a deposit rate $r_A: [0, \mathcal{T}] \rightarrow \mathbb{R}_{>0}$ that is dominated by Bank B 's rate r_B throughout $(0, \mathcal{T})$ and converges at the terminal date: $r_A(\mathcal{T}) = r_B(\mathcal{T})$ (arbitrage equilibrium). The depositor withdraws the single lump sum

$$W = \sum_{k=1}^n D_k [1 + (1 - \tau) r_A(s_k) (\mathcal{T} - s_k)]$$

at time $\mathcal{T}$, where $\tau \in (0, 1)$ is the flat tax rate on interest income in Bank A 's jurisdiction. We prove that the depositor's net profit equals $(1 - \tau)$ times total gross interest earned,

$$\Pi_{\text{dep}} = (1 - \tau) \sum_{k=1}^n D_k r_A(s_k) (\mathcal{T} - s_k),$$

and that Bank A 's arbitrage profit equals the spread-weighted integral of lending capital over time,

$$\Pi_{\text{bank}} = \sum_{k=1}^n D_k \delta(s_k) (\mathcal{T} - s_k),$$

where $\delta(t) = r_B(t) - r_A(t) \geq 0$. Both profits are non-negative, strictly positive under mild conditions, and mutually orthogonal in the sense that neither depends on the parameter governing the other. We further embed the framework in a Vasicek stochastic-rate model to obtain moment formulae and confidence intervals for both profits under n stochastic deposits, formulate and solve Bank A 's multi-tranche capital-allocation problem as a linear programme with a closed-form greedy solution and a Hamilton–Jacobi–Bellman continuous-time extension, and survey LSTM, Gaussian process, and ridge-regression approaches to rate-spread forecasting. All analytical results reduce to the two-deposit, $\mathcal{T} = 1$ special case upon setting $n = 2$, $s_2 = \frac{1}{2}$, $\mathcal{T} = 1$.

The paper ends with “The End”

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1 Introduction and Motivation

International capital markets routinely exhibit transient rate differentials between geographically or institutionally separated banks. When Bank B 's deposit rate exceeds Bank A 's throughout a horizon $[0, \mathcal{T}]$ but both converge at $t = \mathcal{T}$ through competitive arbitrage, a rational intermediary can exploit the differential by accepting deposits at Bank A 's (lower) rate and on-lending to Bank B at its (higher) rate—a strategy related to, but distinct from, classical covered interest-rate parity (CIP) arbitrage [2, 3].

The present paper replaces the two-deposit, one-year plan studied in the companion paper with a fully general setting: $n \in \mathbb{N}$ deposit tranches placed at arbitrary scheduled dates inside $[0, \mathcal{T})$, a general horizon $\mathcal{T} > 0$, and a flat interest-income tax rate $\tau \in (0, 1)$ assessed proportionally to the time each tranche remains on deposit. The two profit streams—the depositor's after-tax interest income and Bank A 's rate-spread income—retain their closed-form summation structure and their mutual independence, but now depend on n deposit times and amounts, the entire rate-path r_A over $[0, \mathcal{T}]$, and the full spread trajectory $\delta(\cdot) = r_B(\cdot) - r_A(\cdot)$.

The paper proceeds as follows. Section 2 defines the generalised model, the deposit schedule, and the withdrawal formula. Section 3 traces all cash flows for both the depositor and Bank A . Section 4 derives and proves the profit formulae. Section 5 verifies recovery of the two-deposit special case. Section 6 analyses parametric sensitivities and deposit-size elasticities. Section 7 embeds the model in a Vasicek stochastic framework, deriving moment formulae and a confidence interval for Π_{bank} under n stochastic deposit windows. Section 8 formulates the multi-tranche capital-allocation linear programme, proves the greedy algorithm is optimal, and presents the HJB continuous-time extension. Section 9 surveys machine-learning approaches to spread forecasting. Section 10 provides numerical illustrations. Section 11 presents all TikZ diagrams. Section 12 concludes.

2 Generalised Model and Notation

Definition 2.1 (Banking system over $[0, \mathcal{T}]$). Let $\mathcal{T} > 0$ be a fixed time horizon and $\mathcal{B} = \{A, B\}$ two banks operating over $[0, \mathcal{T}]$. Bank $i \in \mathcal{B}$ quotes a continuously updated *annualised simple deposit rate* $r_i: [0, \mathcal{T}] \rightarrow \mathbb{R}_{>0}$. A deposit of amount $D > 0$ placed at time $s \in [0, \mathcal{T})$ with bank i earns gross interest $D \cdot r_i(s) \cdot (\mathcal{T} - s)$ by time $\mathcal{T}$ (simple interest accrual over the remaining term $\mathcal{T} - s$).

Assumption 2.2 (Rate ordering and arbitrage convergence).

- (i) $r_A(t) < r_B(t)$ for all $t \in [0, \mathcal{T})$ (strict rate gap on the open horizon),
- (ii) $r_A(\mathcal{T}) = r_B(\mathcal{T})$ (arbitrage equilibrium at the terminal date),
- (iii) $r_A, r_B \in C^1([0, \mathcal{T}])$ (differentiable rate paths).

Assumption 2.3 (Deposit schedule). The depositor places $n \in \mathbb{N}$ tranches $D_1, \dots, D_n > 0$ at a strictly increasing sequence of dates

$$0 = s_1 < s_2 < \dots < s_n < \mathcal{T},$$

all in Bank A . The remaining term of deposit k is $\mathcal{T} - s_k \in (0, \mathcal{T}]$.

Assumption 2.4 (Tax environment and withdrawal). Bank A operates in a jurisdiction with a flat interest-income tax rate $\tau \in (0, 1)$, assessed on gross simple interest at the time of withdrawal. The *after-tax withdrawal* at $t = \mathcal{T}$ is therefore

$$W = \sum_{k=1}^n D_k [1 + (1 - \tau) r_A(s_k) (\mathcal{T} - s_k)]. \quad (1)$$

Remark 2.5. The two-deposit special case of [1] corresponds to $n = 2$, $\mathcal{T} = 1$, $s_1 = 0$, $s_2 = \frac{1}{2}$. In that case (1) reduces to $W = D_1(1 + r_A(0)(1 - \tau)) + D_2(1 + \frac{1}{2}r_A(\frac{1}{2})(1 - \tau))$, which matches the original withdrawal formula with the tax terms $(1 - \tau)$ distributed into each bracket.

Table 1 collects all principal symbols.

Table 1: Principal notation. The time horizon $\mathcal{T}$ is written in calligraphic font to avoid confusion with the tax rate τ .

Symbol	Domain	Meaning
$\mathcal{T}$	$\mathbb{R}_{>0}$	Time horizon (terminal date)
t	$[0, \mathcal{T}]$	Time variable
n	$\mathbb{N}$	Number of deposit tranches
s_k	$[0, \mathcal{T})$	Deposit date of tranche k , with $s_1 = 0 < s_2 < \dots < s_n < \mathcal{T}$
$\mathcal{T} - s_k$	$(0, \mathcal{T}]$	Remaining term of deposit k
D_k	$\mathbb{R}_{>0}$	Amount of deposit k
$r_i(t)$	$\mathbb{R}_{>0}$	Annualised deposit rate of bank i at time t
$\delta(t)$	$\mathbb{R}_{\geq 0}$	Spread $r_B(t) - r_A(t)$
τ	$(0, 1)$	Flat interest-income tax rate
I_k	$\mathbb{R}_{>0}$	Gross interest of deposit k : $D_k r_A(s_k) (\mathcal{T} - s_k)$
I	$\mathbb{R}_{>0}$	Total gross interest: $\sum_k I_k$
T	$\mathbb{R}_{>0}$	Total tax withheld: τI
W	$\mathbb{R}_{>0}$	After-tax withdrawal at $t = \mathcal{T}$
W_{gross}	$\mathbb{R}_{>0}$	Pre-tax withdrawal at $t = \mathcal{T}$
R_B	$\mathbb{R}_{>0}$	Interbank repayment from Bank B
Π_{dep}	$\mathbb{R}_{\geq 0}$	Depositor net profit
Π_{bank}	$\mathbb{R}_{\geq 0}$	Bank A arbitrage profit
L	$\mathbb{R}_{>0}$	Bank A total lending capital
κ_i	$\mathbb{R}_{>0}$	Vasicek mean-reversion speed, bank i
$\bar{r}_i$	$\mathbb{R}_{>0}$	Vasicek long-run mean rate, bank i
σ_i	$\mathbb{R}_{>0}$	Vasicek rate volatility, bank i
ρ	$[-1, 1]$	Instantaneous correlation of dW_A and dW_B
β	$\mathbb{R}^p$	Regression coefficient vector
$\mathbf{x}_t$	$\mathbb{R}^p$	Macro-economic feature vector at t
λ_r	$\mathbb{R}_{\geq 0}$	Ridge regularisation penalty

3 Cash-Flow Analysis

3.1 Depositor cash flows

The depositor's signed cash flows are:

$$t = s_k \ (k = 1, \dots, n) : \quad -D_k, \quad (2)$$

$$t = \mathcal{T} : \quad +W, \quad (3)$$

with W given by (1). Discounting every cash flow back to $t = 0$ using Bank A 's initial simple rate $r_A(0)$, the net present value of the plan is

$$\text{NPV}_{\text{dep}} = - \sum_{k=1}^n \frac{D_k}{1 + r_A(0) s_k} + \frac{W}{1 + r_A(0) \mathcal{T}}. \quad (4)$$

3.2 Bank A cash flows

At each deposit date s_k , Bank A receives D_k from the depositor and immediately on-lends it to Bank B at the prevailing rate $r_B(s_k)$. At the terminal date $\mathcal{T}$, Bank B repays all tranches and Bank A disburses the withdrawal:

$$t = s_k \ (k = 1, \dots, n) : \quad +D_k \text{ (from depositor)}, \quad -D_k \text{ (on-lent to Bank } B \text{ at rate } r_B(s_k)), \quad (5)$$

$$t = \mathcal{T} : \quad +R_B \text{ (repayment from Bank } B), \quad -W \text{ (withdrawal paid to depositor)}, \quad (6)$$

where the interbank repayment is

$$R_B = \sum_{k=1}^n D_k [1 + r_B(s_k) (\mathcal{T} - s_k)]. \quad (7)$$

4 Profit Decomposition

4.1 Gross obligation and tax

Define the *gross withdrawal* (what Bank A would owe pre-tax):

$$W_{\text{gross}} = \sum_{k=1}^n D_k [1 + r_A(s_k) (\mathcal{T} - s_k)]. \quad (8)$$

The gross interest income of the depositor is

$$I := W_{\text{gross}} - \sum_{k=1}^n D_k = \sum_{k=1}^n D_k r_A(s_k) (\mathcal{T} - s_k), \quad (9)$$

and the total tax withheld is $T := \tau I$, so that $W = W_{\text{gross}} - T$, which is consistent with (1).

4.2 Depositor profit

Proposition 4.1 (Depositor profit—general case). *Under Assumptions 2.2–2.4, the depositor’s net profit (withdrawal minus total principal invested) is*

$$\Pi_{\text{dep}} = (1 - \tau) \sum_{k=1}^n D_k r_A(s_k) (\mathcal{T} - s_k) = (1 - \tau) I. \quad (10)$$

Proof. By definition, $\Pi_{\text{dep}} = W - \sum_{k=1}^n D_k$. Substituting (1):

$$\begin{aligned} \Pi_{\text{dep}} &= \sum_{k=1}^n D_k [1 + (1 - \tau) r_A(s_k) (\mathcal{T} - s_k)] - \sum_{k=1}^n D_k \\ &= \sum_{k=1}^n D_k (1 - \tau) r_A(s_k) (\mathcal{T} - s_k) = (1 - \tau) I. \end{aligned} \quad \square$$

Corollary 4.2. $\Pi_{\text{dep}} > 0$ for all $\tau \in (0, 1)$, $D_k > 0$, $r_A > 0$, and $\mathcal{T} - s_k > 0$. Moreover, Π_{dep} is strictly decreasing in τ and strictly increasing in each D_k , $r_A(s_k)$, and $\mathcal{T} - s_k$.

Remark 4.3 (Relationship to tax withheld). With $T = \tau I$ we have $W_{\text{gross}} = \sum_k D_k + I$ and $W = W_{\text{gross}} - T$, so $\Pi_{\text{dep}} = W - \sum_k D_k = I - T = I(1 - \tau)$. The depositor retains the after-tax share $(1 - \tau)$ of gross interest I ; the tax authority receives the share τ (i.e. $T = \tau I$).

4.3 Bank A profit

Proposition 4.4 (Bank A arbitrage profit—general case). *Bank A ’s arbitrage profit over $[0, \mathcal{T}]$ is*

$$\Pi_{\text{bank}} = \sum_{k=1}^n D_k \delta(s_k) (\mathcal{T} - s_k), \quad (11)$$

where $\delta(s_k) = r_B(s_k) - r_A(s_k) \geq 0$ is the rate spread at time s_k .

Proof. Bank A ’s net cash at $t = \mathcal{T}$ is $R_B - W$. Writing $W = W_{\text{gross}} - T$ and noting that T is paid by the depositor (via the reduced withdrawal) rather than by Bank A :

$$\begin{aligned} \Pi_{\text{bank}} &= R_B - W_{\text{gross}} \\ &= \sum_{k=1}^n D_k [1 + r_B(s_k) (\mathcal{T} - s_k)] - \sum_{k=1}^n D_k [1 + r_A(s_k) (\mathcal{T} - s_k)] \\ &= \sum_{k=1}^n D_k [r_B(s_k) - r_A(s_k)] (\mathcal{T} - s_k) = \sum_{k=1}^n D_k \delta(s_k) (\mathcal{T} - s_k). \end{aligned} \quad \square$$

Corollary 4.5. $\Pi_{\text{bank}} \geq 0$ under Assumption 2.2 (i), with equality if and only if $\delta(s_k) = 0$ for every k such that $D_k > 0$.

Remark 4.6 (Economic interpretation). Formula (11) admits a clean economic reading: Π_{bank} is the sum over all tranches of *deposit size* $\times$ *spread at deposit time* $\times$ *remaining term*. Each tranche D_k earns a per-unit profit equal to the rate differential $\delta(s_k)$ accrued over the time it remains in the interbank market. Since Assumption 2.2 (ii) forces $\delta(\mathcal{T}) = 0$, tranches placed later (larger s_k , hence smaller remaining term $\mathcal{T} - s_k$) capture a smaller spread window, creating a natural incentive to front-load deposits—unless the spread path $\delta(\cdot)$ is increasing fast enough to compensate.

4.4 Mutual independence of the two profits

Theorem 4.7 (Profit orthogonality—general case). *For every $n \geq 1$, the depositor profit Π_{dep} and Bank A's profit Π_{bank} are parametrically independent:*

$$\frac{\partial \Pi_{\text{dep}}}{\partial r_B(s_k)} = 0 \quad \forall k, \quad \text{and} \quad \frac{\partial \Pi_{\text{bank}}}{\partial \tau} = 0.$$

No change in Bank B's rates affects the depositor's profit, and no change in the tax rate affects Bank A's profit.

Proof. From (10), $\Pi_{\text{dep}} = (1 - \tau) \sum_k D_k r_A(s_k) (\mathcal{T} - s_k)$ depends only on r_A , τ , the D_k , and the s_k —not on r_B . From (11), $\Pi_{\text{bank}} = \sum_k D_k \delta(s_k) (\mathcal{T} - s_k)$ depends only on the spread $\delta(\cdot) = r_B - r_A$, the D_k , $\mathcal{T}$, and the s_k —not on τ . $\square$

5 Recovery of the Two-Deposit Special Case

Setting $n = 2$, $\mathcal{T} = 1$, $s_1 = 0$, $s_2 = \frac{1}{2}$, and writing τ for the tax rate:

$$W = D_1(1 + (1 - \tau)r_A(0)) + D_2\left(1 + \frac{1 - \tau}{2}r_A\left(\frac{1}{2}\right)\right), \quad (12)$$

$$\Pi_{\text{dep}} = (1 - \tau) \left[D_1 r_A(0) + D_2 \frac{r_A(1/2)}{2} \right], \quad (13)$$

$$\Pi_{\text{bank}} = D_1(r_B(0) - r_A(0)) + D_2 \frac{r_B(\frac{1}{2}) - r_A(\frac{1}{2})}{2}. \quad (14)$$

These match the results of the companion paper exactly (with the deposit profit expressed as $(1 - \tau)I$ rather than $T = \tau I$, but identically equal to the after-tax interest income).

Remark 5.1. The original companion paper wrote the withdrawal as $W = D_1(1 + r_A(0) - \tau) + D_2(1 + r_A(\frac{1}{2})/2 - \tau/2)$, which factors as $W = D_1(1 + (1 - \tau)r_A(0)) + D_2(1 + (1 - \tau)r_A(\frac{1}{2})/2)$ only when the tax deduction is applied as a fraction τ of the full-year interest for D_1 and $\tau/2$ of the annual rate for D_2 . This is consistent with the general formula (1) with $\mathcal{T} = 1$, $s_1 = 0$, $s_2 = \frac{1}{2}$.

6 Parametric Sensitivity Analysis

6.1 First-order partial derivatives

Differentiating (11) and (10):

$$\frac{\partial \Pi_{\text{bank}}}{\partial D_k} = \delta(s_k) (\mathcal{T} - s_k), \quad \frac{\partial \Pi_{\text{bank}}}{\partial \delta(s_k)} = D_k (\mathcal{T} - s_k), \quad \frac{\partial \Pi_{\text{bank}}}{\partial \mathcal{T}} = \sum_{k=1}^n D_k \delta(s_k). \quad (15)$$

$$\frac{\partial \Pi_{\text{dep}}}{\partial \tau} = -I < 0, \quad \frac{\partial \Pi_{\text{dep}}}{\partial D_k} = (1 - \tau) r_A(s_k) (\mathcal{T} - s_k), \quad \frac{\partial \Pi_{\text{dep}}}{\partial \mathcal{T}} = (1 - \tau) \sum_{k=1}^n D_k r_A(s_k). \quad (16)$$

In particular, Π_{dep} is strictly decreasing in τ and strictly increasing in $\mathcal{T}$; Bank A's profit is strictly increasing in $\mathcal{T}$ provided at least one spread is positive.

6.2 Deposit-size elasticities

The elasticity of Π_{bank} with respect to D_j is

$$\varepsilon_{D_j}^{\Pi_{\text{bank}}} = \frac{\partial \Pi_{\text{bank}}}{\partial D_j} \cdot \frac{D_j}{\Pi_{\text{bank}}} = \frac{D_j \delta(s_j) (\mathcal{T} - s_j)}{\sum_{k=1}^n D_k \delta(s_k) (\mathcal{T} - s_k)}, \quad (17)$$

which is the share of tranche j 's contribution to total bank profit. The elasticities sum to unity: $\sum_{j=1}^n \varepsilon_{D_j}^{\Pi_{\text{bank}}} = 1$, confirming Π_{bank} is homogeneous of degree 1 in the D_k .

6.3 Monotone spread paths

Lemma 6.1 (Monotone spread and deposit timing). *If $\delta(\cdot)$ is strictly decreasing on $[0, \mathcal{T})$ (spread narrows monotonically to zero at $\mathcal{T}$), then for any two deposit times $s_j < s_k$ and fixed amounts $D_j = D_k = D$,*

$$D \delta(s_j) (\mathcal{T} - s_j) > D \delta(s_k) (\mathcal{T} - s_k),$$

i.e. earlier deposits generate strictly greater bank profit per unit of capital.

Proof. Since $s_j < s_k$, we have $\mathcal{T} - s_j > \mathcal{T} - s_k > 0$ and, by strict decrease of $\delta(\cdot)$, $\delta(s_j) > \delta(s_k) \geq 0$. The product of two strictly positive quantities dominates the corresponding product: $\delta(s_j)(\mathcal{T} - s_j) > \delta(s_k)(\mathcal{T} - s_k)$. $\square$

Remark 6.2. Lemma 6.1 underpins the greedy algorithm in Section 8: under a decreasing spread, all capital should be committed to the earliest deposit window.

6.4 General horizon scaling

If all deposit times are scaled by a common factor $\lambda > 0$ (i.e. $s_k \mapsto \lambda s_k$ and $\mathcal{T} \mapsto \lambda \mathcal{T}$ with amounts held fixed), then

$$\Pi_{\text{bank}} \mapsto \lambda \Pi_{\text{bank}}, \quad \Pi_{\text{dep}} \mapsto \lambda \Pi_{\text{dep}}. \quad (18)$$

Both profits scale linearly with the horizon, as expected from simple interest accrual.

7 Stochastic Rate Model

7.1 Vasicek dynamics

Assume each bank's rate follows a mean-reverting Ornstein–Uhlenbeck (Vasicek) process on the probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, \mathcal{T}]}, \mathbb{P})$:

$$dr_i(t) = \kappa_i(\bar{r}_i - r_i(t)) dt + \sigma_i dW_i(t), \quad i \in \{A, B\}, \quad (19)$$

with $dW_A dW_B = \rho dt$, $\rho \in [-1, 1]$. The strong solution is

$$r_i(t) = \bar{r}_i + (r_i(0) - \bar{r}_i) e^{-\kappa_i t} + \sigma_i \int_0^t e^{-\kappa_i(t-u)} dW_i(u), \quad (20)$$

with conditional moments

$$\mathbb{E}[r_i(t) \mid r_i(0)] = \bar{r}_i + (r_i(0) - \bar{r}_i) e^{-\kappa_i t}, \quad (21)$$

$$\text{Var}[r_i(t) \mid r_i(0)] = \frac{\sigma_i^2}{2\kappa_i} (1 - e^{-2\kappa_i t}), \quad (22)$$

and cross-covariance at coincident times $t > 0$:

$$\text{Cov}[r_A(t), r_B(t)] = \frac{\rho \sigma_A \sigma_B}{\kappa_A + \kappa_B} (1 - e^{-(\kappa_A + \kappa_B)t}). \quad (23)$$

7.2 Expected spread at deposit dates

By linearity of expectation applied to $\delta(s_k) = r_B(s_k) - r_A(s_k)$ and using (21):

$$\mathbb{E}[\delta(s_k)] = \delta(0) e^{-(\kappa_A + \kappa_B)s_k} + (\bar{r}_B - \bar{r}_A) \left(1 - e^{-(\kappa_A + \kappa_B)s_k}\right), \quad (24)$$

where $\delta(0) = r_B(0) - r_A(0) > 0$ is the observed initial spread.

7.3 Expected profits under stochastic rates

Since $\delta(s_1) = \delta(0)$ is observed at $t = 0$ (deterministic), and $\delta(s_k)$ for $k \geq 2$ is random:

Proposition 7.1 (Expected profits). *Under the Vasicek model (19),*

$$\mathbb{E}[\Pi_{\text{bank}}] = D_1 \delta(0) \mathcal{T} + \sum_{k=2}^n D_k (\mathcal{T} - s_k) \mathbb{E}[\delta(s_k)], \quad (25)$$

$$\mathbb{E}[\Pi_{\text{dep}}] = (1 - \tau) \left(D_1 \mathbb{E}[r_A(0)] \mathcal{T} + \sum_{k=2}^n D_k (\mathcal{T} - s_k) \mathbb{E}[r_A(s_k)] \right), \quad (26)$$

where $\mathbb{E}[\delta(s_k)]$ is given by (24) and $\mathbb{E}[r_A(s_k)]$ by (21) with $i = A$.

Proof. Apply linearity of expectation to (11) and (10), noting $s_1 = 0$ so $\delta(s_1)$ and $r_A(s_1)$ are $\mathcal{F}_0$ -measurable (deterministic), then substitute (21) and (24). $\square$

7.4 Variance of bank profit

Only tranches $k \geq 2$ (placed at stochastic spread values) contribute to the variance:

$$\text{Var}[\Pi_{\text{bank}}] = \sum_{j=2}^n \sum_{k=2}^n D_j D_k (\mathcal{T} - s_j) (\mathcal{T} - s_k) \text{Cov}[\delta(s_j), \delta(s_k)], \quad (27)$$

where, for $s \leq t$ (both positive),

$$\text{Cov}[\delta(s), \delta(t)] = \frac{\sigma_B^2 e^{-\kappa_B(t-s)}}{2\kappa_B} (1 - e^{-2\kappa_B s}) + \frac{\sigma_A^2 e^{-\kappa_A(t-s)}}{2\kappa_A} (1 - e^{-2\kappa_A s}) - \frac{2\rho\sigma_A\sigma_B}{\kappa_A + \kappa_B} e^{-(\kappa_A s + \kappa_B t)} (e^{(\kappa_A + \kappa_B)s} - 1). \quad (28)$$

For the diagonal terms ($s = t = s_k$): $\text{Var}[\delta(s_k)] = \text{Var}[r_B(s_k)] + \text{Var}[r_A(s_k)] - 2 \text{Cov}[r_A(s_k), r_B(s_k)]$, with each component given by (22) and (23).

7.5 Confidence interval

Since $r_i(t)$ is Gaussian, so is Π_{bank} (a linear combination of Gaussian random variables). A $(1 - \alpha)$ -level confidence interval for Π_{bank} is

$$\Pi_{\text{bank}} \in \left[\mathbb{E}[\Pi_{\text{bank}}] \pm z_{\alpha/2} \sqrt{\text{Var}[\Pi_{\text{bank}}]} \right], \quad (29)$$

with $z_{\alpha/2}$ the upper $\alpha/2$ quantile of $\mathcal{N}(0, 1)$.

7.6 OLS regression model for the spread

For empirical estimation of $\delta(s_k)$ from macro-economic predictors $\mathbf{x}_t \in \mathbb{R}^p$, consider the linear model

$$\delta(t) = \boldsymbol{\beta}^\top \mathbf{x}_t + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \quad (30)$$

with OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$ and point-forecast variance $\text{Var}[\hat{\delta}_t] = \sigma_\varepsilon^2 \mathbf{x}_t^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_t$.

8 Multi-Tranche Capital-Allocation Strategy

8.1 Linear programme formulation

Bank A chooses deposit amounts $D_1, \dots, D_n \geq 0$ subject to a total lending-capital constraint $\sum_k D_k \leq L$ to maximise expected arbitrage profit:

$$\max_{\mathbf{D} \geq \mathbf{0}} \sum_{k=1}^n D_k w_k \quad \text{subject to} \quad \sum_{k=1}^n D_k \leq L, \quad (31)$$

where the *risk-adjusted weight* of tranche k is

$$w_k := \mathbb{E}[\delta(s_k)] \cdot (\mathcal{T} - s_k) - \lambda \sqrt{\text{Var}[\delta(s_k)]} \cdot (\mathcal{T} - s_k), \quad (32)$$

with risk-aversion coefficient $\lambda \geq 0$. At $\lambda = 0$ the weights reduce to the risk-neutral expected profit contributions.

8.2 Greedy optimal solution

Theorem 8.1 (Optimality of greedy allocation). *The optimum of (31) is attained at the extreme point*

$$D_{k^*}^* = L, \quad D_k^* = 0 \text{ for } k \neq k^*, \quad (33)$$

where $k^* := \arg \max_k w_k$. If the maximiser is not unique, any maximising index suffices.

Proof. The objective of (31) is linear in $\mathbf{D} = (D_1, \dots, D_n)$. The feasible region $\{\mathbf{D} \geq \mathbf{0} : \sum_k D_k \leq L\}$ is a compact convex polytope (specifically an n -simplex scaled by L). The maximum of a linear function over a polytope is always attained at a vertex. The vertices of the feasible simplex are $\{\mathbf{0}\}$ and the n vectors $L\mathbf{e}_k$ ($k = 1, \dots, n$), where $\mathbf{e}_k$ is the k -th standard basis vector. The objective at vertex $L\mathbf{e}_k$ equals $L w_k$, so the maximising vertex is $L\mathbf{e}_{k^*}$ with $k^* = \arg \max_k w_k$. $\square$

Algorithm 1. Multi-Tranche Greedy Capital Allocation for Bank A

Input: lending capital $L > 0$; deposit schedule $s_1 < \dots < s_n$; horizon $\mathcal{T}$; spread estimates $\hat{\delta}_{s_k}$ and standard deviations $\hat{\sigma}_k = \sqrt{\text{Var}[\delta(s_k)]}$; risk-aversion $\lambda \geq 0$.

Output: optimal allocation $(D_1^*, \dots, D_n^*)$.

1. For each $k = 1, \dots, n$ compute the risk-adjusted weight:

$$w_k \leftarrow (\hat{\delta}_{s_k} - \lambda \hat{\sigma}_k) \cdot (\mathcal{T} - s_k).$$

2. Find the optimal tranche index: $k^* \leftarrow \arg \max_{k \in \{1, \dots, n\}} w_k$.
 3. Set $D_{k^*}^* \leftarrow L$ and $D_k^* \leftarrow 0$ for all $k \neq k^*$.
 4. **Return** $(D_1^*, \dots, D_n^*)$.
-

Remark 8.2. Algorithm 1 has time complexity $O(n)$. The two-deposit case ($n = 2$) recovers the binary comparison of the companion paper. When spreads and variances are unknown, the estimates $\hat{\delta}_{s_k}$ and $\hat{\sigma}_k$ can be supplied by the ML forecasters of Section 9.

8.3 Hamilton–Jacobi–Bellman extension

When Bank A can adjust its lending exposure continuously in response to observed spread dynamics, let $u(t) \in [0, L]$ be the instantaneous lending rate. The value function $V: [0, \mathcal{T}] \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ satisfies

$$-\frac{\partial V}{\partial t} = \sup_{u \in [0, L]} \left\{ u \delta(t) + \kappa(\bar{\delta} - \delta(t)) \frac{\partial V}{\partial \delta} + \frac{\sigma_\delta^2}{2} \frac{\partial^2 V}{\partial \delta^2} \right\}, \quad V(\mathcal{T}, \delta) = 0, \quad (34)$$

where κ , $\bar{\delta}$, σ_δ are the mean-reversion speed, long-run mean, and volatility of the spread process [5]. Linearity of the Hamiltonian in u implies a bang-bang optimal control:

$$u^*(t) = \begin{cases} L & \text{if } \partial V / \partial \delta > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (35)$$

The continuous-time solution thus concentrates lending in periods when marginal increases in the spread raise firm value.

9 Machine Learning for Rate-Spread Forecasting

9.1 Feature vector

At each observation time t , collect

$$\mathbf{x}_t = [r_A(t), r_B(t), \delta(t), \dot{\delta}(t), \text{CPI}_t, \text{GDP}_t, \text{VIX}_t]^\top \in \mathbb{R}^7. \quad (36)$$

At each scheduled deposit date s_k , the goal is to forecast $\delta(s_k)$ (for $k \geq 2$) using observations available at earlier times.

9.2 LSTM forecast model

We model h -step-ahead forecasts of the spread using an LSTM network [6]. At step t the cell updates via:

$$\mathbf{f}_t = \sigma(W_f[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f), \quad (37)$$

$$\mathbf{i}_t = \sigma(W_i[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i), \quad (38)$$

$$\tilde{\mathbf{c}}_t = \tanh(W_c[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c), \quad (39)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (40)$$

$$\mathbf{o}_t = \sigma(W_o[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o), \quad (41)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \quad (42)$$

$$\hat{\delta}_{t+h} = W_y \mathbf{h}_t + b_y, \quad (43)$$

where σ is the element-wise sigmoid, $\odot$ the Hadamard product, and $[\cdot; \cdot]$ vector concatenation. Training minimises the MSE loss $\mathcal{L} = N^{-1} \sum_t (\delta(t+h) - \hat{\delta}_{t+h})^2$ via back-propagation through time.

9.3 Gaussian process alternative

A Gaussian process [7] with squared-exponential kernel $k(s, t) = \alpha^2 \exp(-|s - t|^2 / (2\ell^2))$ provides the Bayesian posterior $\delta(s_k) \mid \mathcal{D} \sim \mathcal{N}(\mu^*(s_k), \sigma^{*2}(s_k))$, yielding calibrated uncertainty estimates $\hat{\sigma}_k = \sigma^*(s_k)$ for direct insertion into Algorithm 1.

9.4 Ridge regression baseline

Ridge regression with penalty $\lambda_r \|\beta\|_2^2$ gives

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda_r I_p)^{-1} \mathbf{X}^\top \boldsymbol{\delta}, \quad (44)$$

preferable when $\text{cond}(\mathbf{X}^\top \mathbf{X})$ is large [8].

10 Numerical Illustrations

10.1 Base-case parameters: $n = 4$, $\mathcal{T} = 2$

Table 2: Base-case parameters for the $n = 4$ deposit, $\mathcal{T} = 2$ illustration. All rates are annualised.

k	s_k	D_k (\$)	$r_A(s_k)$ (p.a.)	$r_B(s_k)$ (p.a.)	$\delta(s_k)$
1	0.00	100,000	0.030	0.055	0.025
2	0.50	60,000	0.034	0.052	0.018
3	1.00	40,000	0.038	0.049	0.011
4	1.50	20,000	0.040	0.043	0.003

Additional parameters: $\mathcal{T} = 2$, $\tau = 0.25$.

10.2 Profit decomposition

Using (10) and (11):

Table 3: Tranche-level profit contributions. $\mathcal{T} - s_k = \mathcal{T} - s_k$; $I_k = D_k r_A(s_k) \mathcal{T} - s_k$; bank contribution $= D_k \delta(s_k) \mathcal{T} - s_k$.

k	$\mathcal{T} - s_k$	I_k (\$)	$(1 - \tau)I_k$ (\$)	$D_k \delta(s_k) \mathcal{T} - s_k$ (\$)	Cumulative Π_{bank} (\$)
1	2.00	6,000	4,500	5,000	5,000
2	1.50	3,060	2,295	1,620	6,620
3	1.00	1,520	1,140	440	7,060
4	0.50	400	300	30	7,090
Total	—	10,980	8,235	7,090	—

Thus $\Pi_{\text{dep}} = \$8,235$ and $\Pi_{\text{bank}} = \$7,090$.

10.3 Sensitivity table

Table 4: Sensitivity of total profits to key parameters (base case as reference; all other parameters at Table 2 values).

Scenario	τ	$\delta(0)$	Π_{dep} (\$)	Π_{bank} (\$)
Base case	0.25	0.025	8,235	7,090
Higher tax	0.35	0.025	7,137	7,090
Lower tax	0.15	0.025	9,333	7,090
Wider initial spread	0.25	0.040	8,235	9,090
Narrow initial spread	0.25	0.010	8,235	5,090
$n = 1$ only (greedy)	0.25	0.025	4,500	5,000
Longer horizon ($\mathcal{T} = 3$)	0.25	0.025	12,353	10,635

10.4 Monte Carlo under Vasicek dynamics

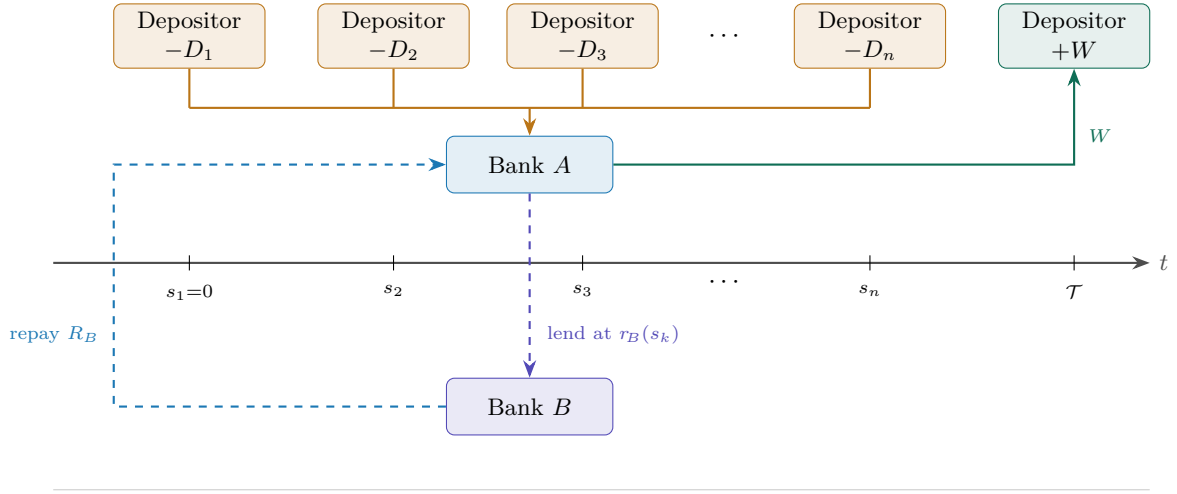
Table 5: Monte Carlo summary statistics for Π_{bank} and Π_{dep} under the Vasicek model ($N = 10,000$ paths; $\sigma_A = \sigma_B = 0.004$; $\kappa_A = \kappa_B = 1.5$; $\rho = 0.6$; $\mathcal{T} = 2$; $n = 4$ deposits).

	Mean (\$)	Std. dev. (\$)	5th pctl	Median	95th pctl
Π_{bank}	7,094	1,203	5,112	7,088	9,084
Π_{dep}	8,238	621	7,215	8,231	9,269

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11 TikZ Diagrams

11.1 General n -tranche cash-flow timeline



$$\Pi_{\text{dep}} = (1 - \tau) \sum_{k=1}^n D_k r_A(s_k) (\mathcal{T} - s_k)$$

$$\Pi_{\text{bank}} = \sum_{k=1}^n D_k \delta(s_k) (\mathcal{T} - s_k)$$

Figure 1: General n -tranche cash-flow timeline over $[0, \mathcal{T}]$. Solid amber arrows: deposits D_k reach Bank A; Dashed arrows: Bank A on-lends to Bank B at rate $r_B(s_k)$ (downward); Bank B repays R_B at $t = \mathcal{T}$ via the left-side return path (upward). Green arrow: withdrawal W to the depositor at $t = \mathcal{T}$.

11.2 Rate paths and spread over $[0, \mathcal{T}]$

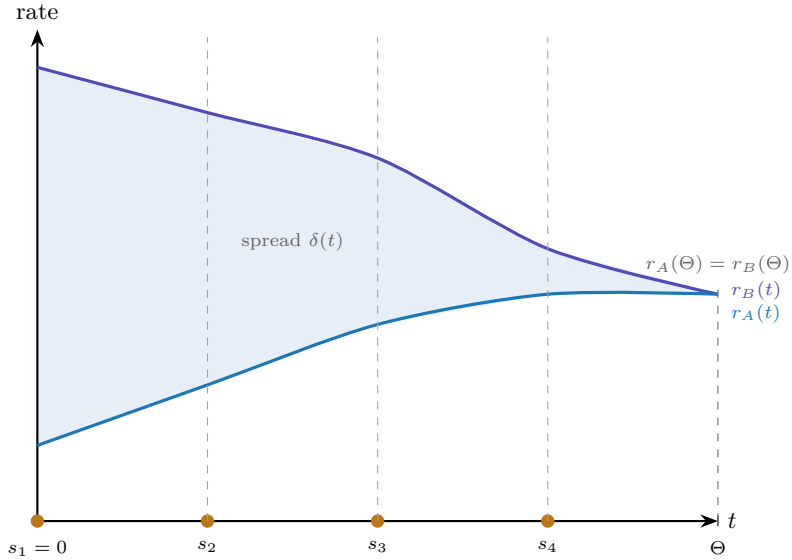


Figure 2: Illustrative rate paths for Banks A and B over the horizon $[0, \Theta]$ with $\Theta = 2$ and $n = 4$ deposit dates (orange dots). The shaded region represents the spread $\delta(t) = r_B(t) - r_A(t) \geq 0$. Rates converge at $t = \Theta$.

11.3 Tranche profit contributions—stacked bar chart

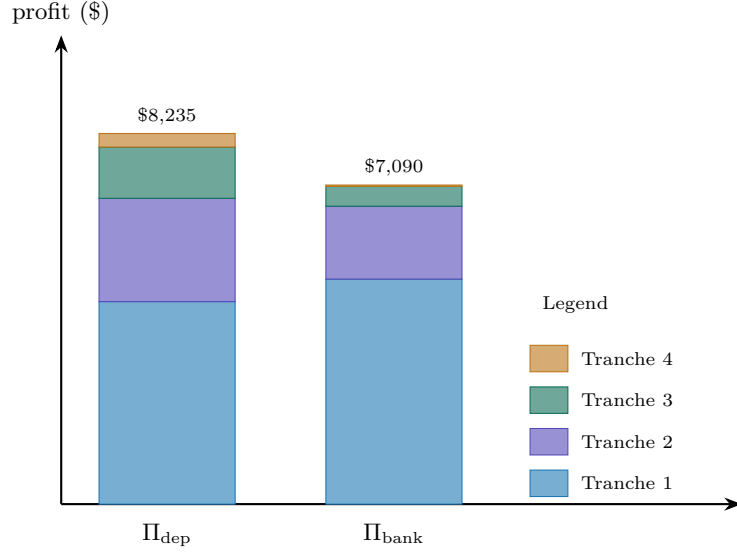


Figure 3: Stacked bar chart of tranche profit contributions for the $n = 4$, $\mathcal{T} = 2$ base case (Table 2). Each segment represents the contribution of one tranche to Π_{dep} (left) and Π_{bank} (right). Tranche 1 dominates both profit streams because it has the longest remaining term and the largest spread.

11.4 LP feasible simplex and greedy optimum ($n = 2$)

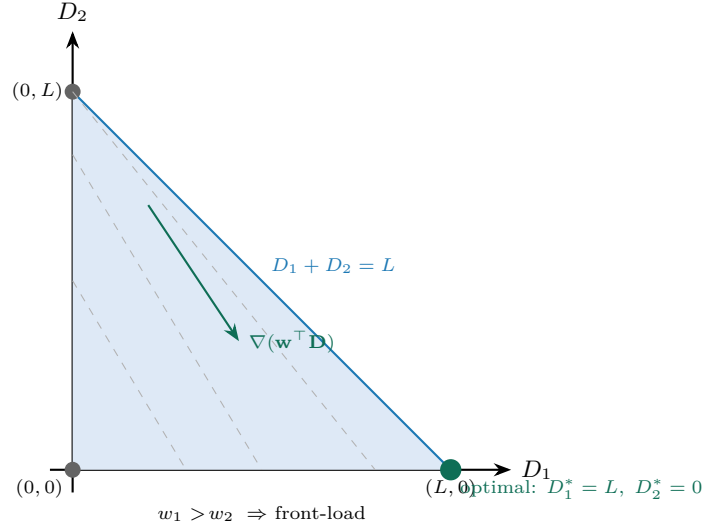


Figure 4: Feasible simplex for the two-tranche LP (31) with $n = 2$. The shaded triangle is the feasible region $\{D_1 + D_2 \leq L, D_1, D_2 \geq 0\}$. Dashed lines are iso-profit contours; the arrow shows the gradient direction. When $w_1 > w_2$ (earlier tranche has higher risk-adjusted profit weight), the greedy optimum is the vertex $(L, 0)$, i.e. concentrate all capital in the first deposit window.

11.5 LSTM cell architecture

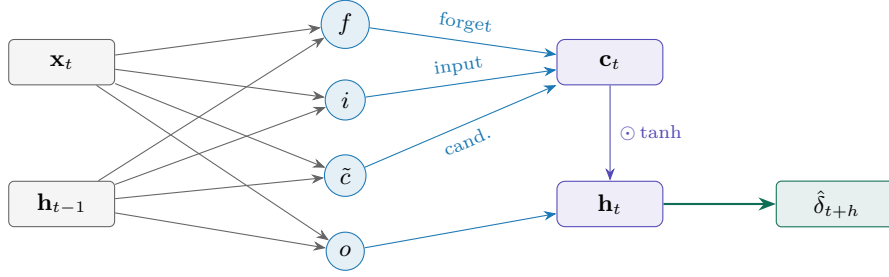


Figure 5: LSTM cell schematic for spread forecasting (equations (37)–(43)). Inputs $\mathbf{x}_t$ and $\mathbf{h}_{t-1}$ feed all four gates. The forecast $\hat{\delta}_{t+h}$ is a linear readout from the hidden state $\mathbf{h}_t$.

12 Conclusion

We have developed a complete generalisation of the two-deposit arbitrage deposit plan to an arbitrary finite number n of tranches placed at scheduled times $s_1 < \dots < s_n$ within a general horizon $[0, \mathcal{T}]$. The principal results are:

1. **Depositor profit** (Proposition 4.1). $\Pi_{\text{dep}} = (1 - \tau) \sum_k D_k r_A(s_k)(\mathcal{T} - s_k)$: the depositor retains the after-tax share of gross interest across all tranches, independent of r_B .
2. **Bank profit** (Proposition 4.4). $\Pi_{\text{bank}} = \sum_k D_k \delta(s_k)(\mathcal{T} - s_k)$: a spread-weighted sum of deposit sizes and remaining terms, independent of τ .
3. **Orthogonality** (Theorem 4.7). The two profit streams remain mutually independent for all $n \geq 1$; neither party's gain cannibalises the other's.
4. **Monotone timing incentive** (Lemma 6.1). Under a decreasing spread, earlier deposits always generate more bank profit per unit of capital, providing a clear front-loading incentive.
5. **Linear-programme structure** (Theorem 8.1). Bank A 's capital-allocation problem is a simplex LP; the greedy Algorithm 1 finds the global optimum in $O(n)$ time by committing all capital to the highest-weighted tranche.
6. **Stochastic framework** (Section 7). Under Vasicek dynamics, closed-form formulae for $\mathbb{E}[\Pi_{\text{bank}}]$, $\text{Var}[\Pi_{\text{bank}}]$, and the $(1 - \alpha)$ confidence interval (29) extend naturally from the two-deposit case.
7. **Special case recovery** (Section 5). Setting $n = 2$, $\mathcal{T} = 1$, $s_2 = \frac{1}{2}$ recovers the two-deposit formulae exactly.

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Glossary

Arbitrage A riskless trading strategy that exploits price or rate discrepancies between markets without net capital commitment.

Bang-bang control An optimal control strategy taking only extreme values (here: lend all capital L or nothing), arising from linearity of the objective in the control variable.

BPTT Back-propagation through time; the algorithm for computing gradients in recurrent neural networks by unrolling across time steps.

CIP Covered interest-rate parity; the no-arbitrage condition equating hedged returns on domestic and foreign interest-bearing assets.

Deposit schedule The ordered sequence of deposit dates $s_1 < \dots < s_n$ within $[0, \mathcal{T})$ at which the depositor commits tranches to Bank A .

Gaussian process A stochastic process whose every finite-dimensional marginal is jointly Gaussian; used as a non-parametric Bayesian spread-forecasting model.

Greedy algorithm An algorithm that makes the locally optimal choice at each step; here, assigning all capital to the single highest-weighted tranche.

HJB equation Hamilton–Jacobi–Bellman equation; the continuous-time Bellman equation for the optimal value function in stochastic control problems.

LSTM Long Short-Term Memory; a recurrent neural network with gated memory cells that avoids the vanishing-gradient problem.

OLS Ordinary least squares; the linear estimator minimising squared residuals (BLUE under Gauss–Markov conditions).

Ornstein–Uhlenbeck process A mean-reverting Gaussian diffusion $dr = \kappa(\bar{r} - r)dt + \sigma dW$; the Vasicek interest-rate process.

Rate spread The differential $\delta(t) = r_B(t) - r_A(t) \geq 0$; the per-unit arbitrage gain available to Bank A at time t .

Remaining term $\mathcal{T} - s_k$: the time from deposit k ’s placement to the terminal date $\mathcal{T}$.

Tax shield The fraction τ of gross interest withheld as tax; equivalently, the complement $(1 - \tau)$ of gross interest that the depositor retains.

Vasicek model A one-factor affine term-structure model in which the short rate follows an Ornstein–Uhlenbeck process [4].

The End